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-- CREATE STUDENTS AND DEPARTMENTS TABLES
CREATE TABLE Students(
    student_id INT PRIMARY KEY,
    student_name VARCHAR(100) UNIQUE NOT NULL,
    std_dept_name VARCHAR(100)
);

CREATE TABLE Departments(
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(100) UNIQUE NOT NULL
);

-- INSERT 4 RECORDS INTO STUDENTS
INSERT INTO Students VALUES
    (1, 'Ahin Paul', 'CSE'),
    (2, 'Palash Kirtaniya', 'CSE-AI'),
    (3, 'Ahina Sen', 'CSE-DS'),
    (4, 'Kalyan Mondal', NULL);

-- INSERT 4 RECORDS INTO DEPARTMENTS
INSERT INTO Departments VALUES
    (100, 'CSE'),
    (101, 'CSE-AI'),
    (102, 'CSE-DS'),
    (103, 'CSE-CS');

-- CHECK ALL RECORDS
SELECT * FROM Students;
SELECT * FROM Departments;

-- Q1. DISPLAY EACH STUDENT'S NAME TOGETHER WITH
--     THEIR DEPARTMENT ID
SELECT student_name, dept_id
FROM Students s
INNER JOIN Departments d
ON s.std_dept_name = d.dept_name;

-- Q2. DISPLAY THE NAMES OF STUDENTS WHO BELONG
--     TO DEPARTMENT 101
SELECT student_name
FROM Students s
INNER JOIN Departments d
ON s.std_dept_name = d.dept_name
WHERE d.dept_id = 101;

-- Q3. DISPLAY ALL STUDENTS AND THEIR DEPARTMENT ID,
--     INCLUDING STUDENTS WITHOUT A DEPARTMENT
SELECT s.*, d.dept_id
FROM Students s
LEFT JOIN Departments d
ON s.std_dept_name = d.dept_name;

-- Q4. USE AN EQUI JOIN TO DISPLAY STUDENT NAME
--     AND DEPARTMENT
SELECT *
FROM Students s, Departments d
WHERE s.std_dept_name = d.dept_name;

-- Q5. DISPLAY STUDENT NAME AND DEPARTMENT ID
--     WHERE DEPARTMENT ID IS 102 USING EQUI JOIN
SELECT s.student_name, d.dept_id
FROM Students s, Departments d
WHERE s.std_dept_name = d.dept_name
AND d.dept_id = 102;

-- Q6. DISPLAY DEPARTMENT ID ALONG WITH STUDENT NAME,
--     INCLUDING DEPARTMENTS WITH NO STUDENTS
SELECT d.dept_id, s.student_name
FROM Students s
RIGHT JOIN Departments d
ON s.std_dept_name = d.dept_name;

-- Q7. DISPLAY EVERY STUDENT NAME AND DEPARTMENT ID,
--     INCLUDING UNMATCHED COMBINATIONS
SELECT Students.student_name, Departments.dept_id
FROM Students
CROSS JOIN Departments;

-- ALTERNATIVE FOR Q7:
-- DISPLAY ALL STUDENTS AND ALL DEPARTMENTS,
-- INCLUDING UNMATCHED ROWS
SELECT Students.student_name, Departments.dept_id
FROM Students
LEFT JOIN Departments
ON Students.std_dept_name = Departments.dept_name
UNION
SELECT Students.student_name, Departments.dept_id
FROM Students
RIGHT JOIN Departments
ON Students.std_dept_name = Departments.dept_name;

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Result Grid			
student_id	student_name	std_dept_name	
1	Ahin Paul	CSE	
2	Palash Kirtaniya	CSE-AI	
3	Ahina Sen	CSE-DS	
4	Kalyan Mondal	NULL	
5	NULL	NULL	

Result Grid		
dept_id	dept_name	
100	CSE	
101	CSE-AI	
103	CSE-CS	
102	CSE-DS	
NULL	NULL	

Result Grid		
student_name	dept_id	
Ahin Paul	100	
Palash Kirtaniya	101	
Ahina Sen	102	

Result Grid	
student_name	
Palash Kirtaniya	

Result Grid				
student_id	student_name	std_dept_name	dept_id	
1	Ahin Paul	CSE	100	
2	Palash Kirtaniya	CSE-AI	101	
3	Ahina Sen	CSE-DS	102	
4	Kalyan Mondal	NULL	NULL	

Result Grid					
student_id	student_name	std_dept_name	dept_id	dept_name	
1	Ahin Paul	CSE	100	CSE	
2	Palash Kirtaniya	CSE-AI	101	CSE-AI	
3	Ahina Sen	CSE-DS	102	CSE-DS	

Result Grid		
student_name	dept_id	
Ahina Sen	102	

Result Grid	
dept_id	student_name
100	Ahin Paul
101	Palash Kirtaniya
103	NULL
102	Ahina Sen

Result Grid		
student_name	dept_id	
Palash Kirtaniya	100	
Kalyan Mondal	100	
Ahina Sen	100	
Ahin Paul	100	
Palash Kirtaniya	101	
Kalyan Mondal	101	
Ahina Sen	101	
Ahin Paul	101	
Palash Kirtaniya	103	
Kalyan Mondal	103	
Ahina Sen	103	
Ahin Paul	103	
Palash Kirtaniya	102	
Kalyan Mondal	102	
Ahina Sen	102	
Ahin Paul	102	